

Yan Liu

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Education

School of Mechanical Engineering, Shanghai Jiao Tong University (SJTU), China

- Ph.D. Mar. 2026
Advisor: Prof. Xiaojing Liu, Distinguished Professor in Energy
Dissertation: Multiscale coupling of interfacial transport and feedback in complex systems
- B.S. Jun. 2021

Research

Research Focus

- Interfacial Heat and Mass Transfer
- Multiscale Transport in Structured Media
- Interface-Sensitive Measurement and Diagnostics

Research Experience

- (1) **Interfacial phase-change-driven transport phenomena** – Investigated bubble dynamics, microlayer evaporation, and phase-change-driven deposition under high heat-flux and non-equilibrium conditions. Developed an accelerated experimental platform to study coupled heat and mass transfer at solid-liquid-vapor interfaces, achieving strong agreement with benchmark measurements.
- (2) **Micro/nanoscale transport in porous and structured media** – Studied species transport, accumulation, and phase stability in porous and structured media. Demonstrated how micro/nano-scale morphology and internal transport resistance govern heat transfer, mass diffusion, and species redistribution.
- (3) **Physics-based multi-scale modeling of thermal transport systems** – Developed multi-physics transport models integrating heat transfer, mass transport, reaction kinetics, and porous-media physics. Applied model-guided analysis to connect micro/nano-scale mechanisms with device- and system-level performance, enabling predictive assessment of transport-limited processes in aqueous systems.

Additional Research & Training

- **Surface engineering & wettability control:** Bi-thermal-conductivity surfaces and boiling-induced self-assembled micro/nanostructured coatings with experience in surface-driven wetting and transport regulation.
- **Microfluidic & confined-flow exposure:** Involved in exploratory studies of microfluidic transport in supercritical CO₂ systems, developing intuition for microscale flow control and non-equilibrium transport.
- **Advanced characterization:** Hands-on experience with Raman spectroscopy, neutron diffraction, and prompt gamma techniques for interfacial and transport-related measurements.
- **Data-driven & optimization tools:** Experience with topology-based structural optimization, nonlinear system modeling, and physics-informed data analysis to support transport optimization.

Skills

- **Modeling & Simulation & Data Analysis** – Python, C++, CFD, Machine learning
- **Experimental Techniques** – Interfacial heat & mass transfer experiment, Micro/nano-surface fabrication
- **Others** – Thermodynamic-kinetic database development, Image-based morphology and transport analysis

Publications

Full list of publications available on my [personal website](#).

- **Yan Liu**, Xiaojing Liu, Tengfei Zhang & Hui He. Understanding flow fouling deposition and solute hideout-return behavior at the phase change interface. *ACS Applied Materials & Interfaces* 16(4), 4719-4728 (2024). (Q1)
- **Yan Liu**, Xiaojing Liu, Hui He, Tengfei Zhang & Xiang Chai. Synthesizing nuclear power plant fouling with fractal characteristics enables an in-depth study of concerned nuclear safety issues. *iScience* 27(2), 108789 (2024). (Q1)
- **Yan Liu**, Hui He, Tengfei Zhang & Xiaojing Liu. Pressurized water reactor fuel corrosion-related unidentified deposit and its related safety issues - I. Corrosion product deposition and heat transfer. *Annals of Nuclear Energy* 208, 110758 (2024). (Q1)
- **Yan Liu**, Jiajie Chen, Guolian Wang, Hui He, Tengfei Zhang & Xiaojing Liu. Pressurized water reactor fuel corrosion-related unidentified deposit and its related safety issues - II. Corrosion product deposition and heat transfer modeling. *Annals of Nuclear Energy* 211, 110932 (2024). (Q1)
- **Yan Liu**, Guolian Wang, Hui He, Tengfei Zhang & Xiaojing Liu. Pressurized water reactor fuel corrosion-related unidentified deposit and its related safety issues - III. N-TH-M coupled CIPS prediction. *Annals of Nuclear Energy* 208, 110800 (2024). (Q1)
- Chuandong Liu, **Yan Liu**, Hui He, Tengfei Zhang & Xiaojing Liu. Critical heat flux on boiling induced self-assembly surfaces. *International Journal of Heat and Mass Transfer* 257, 128230 (2026). (Q1)
- Guolian Wang, **Yan Liu**, Xiaojing Liu & Hui He. Study on coupling effect of CRUD growth and heat transfer on fuel rod. *International Journal of Thermal Science* 199, 108892 (2024). (Q1)
- Guolian Wang, **Yan Liu**, Xiaojing Liu, Xiang Chai, Tengfei Zhang & Hui He. Study on coupling effect of CRUD growth and heat transfer on fuel rod. *International Journal of Thermal Science* 220(A), 110268 (2026). (Q1)
- Hui He, **Yan Liu**, Shiwei Wang, Tengfei Zhang, Xiang Chai & Xiaojing Liu. Review on corrosion-related unidentified deposit of pressurized water reactors. *Nuclear Engineering and Design* 441, 114095 (2025). (Q1)
- Chaoyuan Zhang, Han Yin, **Yan Liu**, Hui He, Xiaojing Liu, Tengfei Zhang & Sicheng Wang. High-resolution multi-physics analysis of CRUD effects in reactor assemblies. *PRX Energy* 4(3), 033016 (2025).
- **Yan Liu**, Xiaojing Liu, Hui He, Tengfei Zhang & Xiang Chai. Study on thermal conductivity model for porous PWR CRUD depositions. in Proceeding of the 31st International Conference on Nuclear Engineering (ICONE-31) (American Society of Mechanical Engineers, Prague, Czech Republic, 2024). (Winner of Best Paper Award)
- Yan Liu, Xiaojing Liu, Sijia Du, Jiageng Wang & Hui He. Multi-physics coupling model for thermal hydraulics and solute transport in CRUD deposits. in Proceedings of the 23rd Pacific Basin Nuclear Conference (PBNC2022) (Springer Nature, Beijing & Chengdu, China, 2022). (Winner of Best Poster Award)
- Xinyu Li, **Yan Liu**, Xiang Chai & Xiaojing Liu. Dynamic simulation and performance analysis of a nuclear-solar energy system with thermal energy storage. in Proceeding of the 21st International Topical Meeting on Nuclear Reactor Thermal Hydraulics (NURETH-21) (American Nuclear Society, Busan, Korea, 2025). (Winner of Best Poster Award)
- Longchang Tang, Wei Xu, Yan Liu & Xiaojing Liu*. Experimental study on the influence of epoxy patterns of bi-conductive surfaces on pool boiling. in Proceeding of the 21st International Topical Meeting on Nuclear Reactor Thermal Hydraulics (NURETH-21) (American Nuclear Society, Busan, Korea, 2025).
- Chaoyuan Zhang, Han Yin, Yan Liu, Hui He, Xiaojing Liu & Tengfei Zhang*. High-fidelity CRUD coupling simulation for PWR power shift. in Proceeding of the 32nd International Conference on Nuclear Engineering (ICONE-32) (Springer Nature, Weihai, China, 2025). (Winner of Best Poster Award)

Honors & Awards

- 2025 China National Scholarship (Top 0.2%), Ministry of Education of the People's Republic of China
- 2024 China National Scholarship (Top 0.2%), Ministry of Education of the People's Republic of China
- 2022 China National Scholarship (Top 0.2%), Ministry of Education of the People's Republic of China
- 2025 Special Talent Cultivation Project for PhD Students, The China Association for Science and Technology
- Ph.D. Graduate with Distinction: Excellence Award of Shanghai, Shanghai Municipal Education Commission
- 2024 Best Paper Award, The 31st International Conference on Nuclear Engineering ICONE-31
- 2022 Best Poster Award, The 23rd Pacific Basin Nuclear Conference PBNC2022
- 2024 ENEN++ Nuclear Mobility Fund, European Nuclear Education Network ENEN
- 2023 Outstanding Postgraduate Presentation Award, Shanghai Nuclear Society
- 2023 President's Award of SJTU, Shanghai Jiao Tong University

Professional Services & Memberships

- **Conference Organization** – Section Co-chair, The 31st International Conference on Nuclear Engineering
- **Academic Advising** – Faculty Advisor for B.S. Nuclear Class of 2025, Shanghai Jiao Tong University
- **Professional Societies** –Member, American Nuclear Society; American Society of Mechanical Engineers

References

- Prof. Xiaojing Liu (Ph.D. Advisor)
Executive Vice Dean of College of Smart Energy, Shanghai Jiao Tong University
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